

Issac Trotts

Senior Software Engineer · Go · Distributed Systems

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Colorado · Remote (US)

SUMMARY

Senior engineer with 20+ years of shipping production software: abuse prevention at YouTube scale, the portable Pub/Sub API in Google's Go Cloud, the first PHP 7 runtime on App Engine, a rewritten NLP engine for an AI trading system, and a mobile app for offline trip itineraries. I work well remotely and independently and am in the habit of verifying things.

SKILLS

Languages	Go · Elixir · Python · C++ · Dart · Rust · SQL
Platforms	Phoenix LiveView · Postgres · Google Cloud / App Engine · Pub/Sub · Qt · Flutter (iOS & Android)
Strengths	Visualization · Distributed systems · Performance tuning · Abuse & spam detection · NLP · Automated testing

EXPERIENCE

illumineX — Software Engineering Contractor

Nov 2021 – Present · Remote

- Rewrote the core **NLP engine** for the mistro AI trading system, significantly improving accuracy and throughput, and added 1,000+ automated tests.
- Built **first.then.next**, an AI-powered calendar focused on user independence and goal achievement.

Stairwell — Software Engineer

Apr 2021 – Oct 2021 · Remote

- Built a new malware ingestion pipeline, expanding cloud-scale threat intelligence and detection.

Speakeasy — Cofounder & Engineer

Apr 2020 – Apr 2021 · Remote

- Architected a full-stack video events platform in **Elixir with Phoenix LiveView** that reliably supported **300+ concurrent participants**.
- Landed the first paying customer with my cofounders and ran recurring high-attendance weekly events.

YouLi — Mobile App Contractor

2019 – 2020 · Remote

- Built the **YouLi Trip** app for iOS and Android with full offline support for trip pages.
- No reported crashes and only cosmetic maintenance in its first six years in production.

Google — Software Engineer

Apr 2016 – Mar 2019 · San Francisco

Go Cloud · Aug 2017 – Mar 2019

- Designed and shipped the portable **Pub/Sub API** (gocloud.dev/pubsub), letting Go services send and receive messages across GCP, AWS, Azure, Kafka, NATS and more through one interface.

App Engine Runtimes · Apr 2016 – Aug 2017

- Launched the **PHP 7.2 runtime** (beta), which was widely adopted and hit #1 on Hacker News.

VideoSurf — Software Engineer

Oct 2010 – Sep 2012 · San Mateo

- Sped up the text-mining and deduplication pipeline from seconds to **50 ms** per video (~70×).
- Used Postgres indexing to make catalog updates thousands of times faster, enabling daily full-site refreshes.
- Built fuzzy-matching daemons for duplicate video detection, plus automated data-quality reports.

- Built a Sawzall-based **abuse detection** suite with stealth countermeasures that neutralized mass spam accounts without tipping off attackers.
- Shut down the chain-comment exploit, added spam filters, and built brute-force password attack mitigations.
- Built dependency tracking for denormalized views in a sharded data warehouse, with atomic updates across shards.

OPEN SOURCE

Ohm

A minimal web browser for Omarchy Linux: one window per page, with Hyprland as the tab bar. Warm opens in under 150 ms.

C++ · Qt WebEngine · 2026

Go Cloud Pub/Sub

Portable, driver-based pub/sub API for Go. Designed and built at Google.

Go · distributed messaging

go-anytime

Parses natural and standard dates, times and ranges in Go without knowing the format in advance.

Go · parser combinators

cmakelists_parsing · trigram

A Python parser for CMakeLists.txt, and trigram-based string similarity in Rust.

Python · Rust

EARLIER

- **Google, Ad Spam Operations** — Built a phone verification system that cut failures to under 25% of the old system's, plus a GWT UI for spam investigations. 2007–2008
- **UC Davis** — Programmer IV. Built 3D brain visualization tools. 2002–2006
- **Idol Minds** — Adapted mesh simplification algorithms for PS2 game rendering. 2001–2002
- **NASA Ames** — Intern. Implemented tetrahedral mesh simplification with scalar fields. 1999

EDUCATION

Brown University

M.Sc., Applied Mathematics

University of California, Davis

B.S., Mathematics